

Methodology of the PlasMol Fourier Driver

Including Meep Vacuum Field References for Hybrid Spectra

PlasMol Development Notes

July 20, 2026

Contents

1	Purpose and scope	2
2	Physical overview	2
2.1	Induced dipole from RT-TDDFT	2
2.2	From time series to absorption	2
3	Common post-processing pipeline	2
3.1	Folding diagonal response components	3
3.2	Optional exponential damping	3
3.3	Discrete Fourier transform	3
3.4	Outputs	3
4	Quantum-only Fourier path (delta kicks)	3
4.1	Setup	3
4.2	Typical JSON fragment	4
5	Hybrid Meep Fourier path	5
5.1	Motivation	5
5.2	Production runs (molecule + NP)	5
5.3	Unit and time-grid notes	5
5.4	Spectral deconvolution	5
5.4.1	Consistent deconvolution	6
5.4.2	Complex division, phase cancellation, and the amplitude floor	7
5.4.3	From α_i to the reported absorption	8
5.4.4	Compact algorithm (as implemented)	8
5.5	Example hybrid JSON (Fourier + Au NP)	8

1 Purpose and scope

The **Fourier driver** is the recommended workflow for computing linear absorption spectra. Given a short, broadband excitation, the code:

1. runs three independent polarizations (x, y, z) in parallel;
2. records the induced molecular dipole $\boldsymbol{\mu}(t)$;
3. optionally damps the time-domain signals to artificially increase the likelihood of convergence;
4. Fourier-transforms the response;
5. assembles a normalized absorption spectrum over a user-chosen energy window.

2 Physical overview

2.1 Induced dipole from RT-TDDFT

At each quantum step the molecule is propagated under the instantaneous electric field $\mathbf{E}(t)$ (either an analytic kick/pulse or the local Meep field). The induced dipole is

$$\mu_i(t) = \text{Tr}[\hat{\mu}_i(\mathbf{D}(t) - \mathbf{D}_0)], \quad i \in \{x, y, z\}, \quad (1)$$

where $\mathbf{D}(t)$ is the time-dependent density matrix in the AO basis, $\mathbf{D}_0$ is the ground-state density, and $\hat{\mu}_i$ are dipole integrals (`molecule.calculate_mu()`). Open-shell runs sum α and β blocks before the trace.

2.2 From time series to absorption

PlasMol builds a scalar absorption-like quantity from the imaginary part of that response along all three Cartesian directions and a prefactor proportional to frequency:

$$A(\omega) = -\frac{4\pi\omega}{3c} \sum_{i \in \{x, y, z\}} \text{Im}[\mathcal{R}_i(\omega)], \quad (2)$$

where c is the speed of light in atomic units (`constants.C_AU`) and $\mathcal{R}_i(\omega)$ is either the Fourier-transformed dipole or the deconvolved ratio $\mu_i(\omega)/E_{\text{inc},ii}(\omega)$ depending on the source kick; see Sections 4–5.4. The published curve is peak-normalized:

$$A_{\text{norm}}(\omega) = \frac{A(\omega)}{\max_{\omega} |A(\omega)|}. \quad (3)$$

Frequencies are obtained from the real FFT frequency grid,

$$\omega_k = 2\pi f_k, \quad f_k = \text{fftfreq}(N, \Delta t), \quad (4)$$

converted to eV via $E[\text{eV}] = \omega[\text{a.u.}] \times 27.211386$, and restricted to the user window `[min_ev, max_ev]`.

3 Common post-processing pipeline

Regardless of how the directional runs are obtained (Quantum only or Full Plasmol), the post-processing stage is shared.

3.1 Folding diagonal response components

Each directional run writes a full Cartesian field CSV. Only the component parallel to the drive is retained and stacked into a $(3 \times N)$ array:

$$\boldsymbol{\mu}_{\text{diag}}(t) = (\mu_x^{(x)}(t), \mu_y^{(y)}(t), \mu_z^{(z)}(t))^{\top} \quad (\text{function fold}). \quad (5)$$

Here the superscript denotes the polarization of the incident source. Time grids are required to share a common Δt ; series are trimmed to the shortest length if they differ slightly.

3.2 Optional exponential damping

Two related damping mechanisms may be applied before the FFT.

γ window. Inside `fourier()`, every time series is multiplied by

$$w_{\gamma}(t) = e^{-\gamma t}, \quad (6)$$

with $\gamma = \text{fourier.gamma}$ (a.u.; default = 0). This broadens spectral features artificially.

τ damping. If `fourier.tau` > 0, an additional factor

$$w_{\tau}(t) = e^{-t/\tau} \quad (7)$$

is applied to the polarizability CSVs on disk (written as `*_damped.csv`) and, for the Meep path, to the reference $\mathbf{E}_{\text{inc}}$ in memory only so that the reusable reference file remains undamped.

3.3 Discrete Fourier transform

For each Cartesian axis i ,

$$S_{\mu,i}(\omega) = \Delta t \sum_{n=0}^{N-1} \mu_i(t_n) w(t_n) e^{-i\omega t_n}, \quad (8)$$

implemented as `np.fft.fft(...)` * `dt`. The same window w multiplies $\mathbf{E}_{\text{inc}}$ when deconvolution is active.

3.4 Outputs

- `spectrum_filepath` — PNG of $A_{\text{norm}}(E)$;
- sibling `.csv` with columns `Frequency`, `Absorption`;
- optional `npz_filepath` storing `abs_imag`, `abs_real`, `freqs`, and a `deconvolved` flag;
- per-direction working directories `x_dir/`, `y_dir/`, `z_dir/` with raw `field_e.csv` / `field_p.csv` (and vacuum reference CSVs when applicable).

4 Quantum-only Fourier path (delta kicks)

4.1 Setup

When the input has a `molecule` section but no `plasmon` section, the Fourier driver schedules three pure RT-TDDFT jobs. For each direction $i \in \{x, y, z\}$:

- `molecule.source.component` is forced to i ;

- QUANTUMSOURCE builds an analytic field $\mathbf{E}^{(i)}(t)$ on the global time grid $t_n = n \Delta t$;
- for `type:"kick"`, the active component is a one-step δ -like impulse of strength `intensity` at `peak_time`.

A true δ -kick has a flat spectrum, so the polarizability is proportional to the Fourier transform of the induced dipole itself. The driver therefore calls

$$\mathcal{R}_i(\omega) = S_{\mu,i}(\omega) \quad (\text{field_e} = \text{None branch of } \text{fourier}()), \quad (9)$$

i.e. $\text{Im}[\mu_i(\omega)]$ enters Eq. (2) directly.

4.2 Typical JSON fragment

```
{
  "settings": {
    "dt": 0.1,
    "t_end": 4000,
    "driver": "fourier"
  },
  "molecule": {
    "geometry": [
      {"atom": "O", "coord": [ 0.000, 0.000, -0.130]},
      {"atom": "H", "coord": [ 1.489, 0.000,  1.033]},
      {"atom": "H", "coord": [-1.489, 0.000,  1.033]}
    ],
    "geometry_units": "bohr",
    "basis": "631g*",
    "xc": "pbe0",
    "source": {
      "type": "kick",
      "intensity": 5e-5,
      "peak_time": 0.05,
      "width_steps": 3
    }
  },
  "files": {
    "checkpoint": {
      "filepath": "checkpoint.npz",
      "frequency_steps": 1000
    },
    "field_e_filepath": "field_e.csv",
    "field_p_filepath": "field_p.csv",
    "spectra_e_vs_p_filepath": "output.png"
  },
  "additional_parameters": {
    "fourier": {
      "min_ev": 1,
      "max_ev": 35,
      "npz_filepath": "fourier.npz",
      "spectrum_filepath": "spectrum.png"
    }
  }
}
```

5 Hybrid Meep Fourier path

5.1 Motivation

When a nanoparticle (or any classical structure) is present, the molecule no longer sits in a free-space δ -kick. Instead Meep propagates a broadband source (typically a **gaussian** pulse) through the FDTD grid; the local field at `plasmon.molecule.position` drives RT-TDDFT, and the induced dipole may be fed back into Meep as a point `CustomSource` (`back_propagation`). Two complications follow:

1. the incident pulse has a non-flat spectrum and a non-trivial phase;
2. the local field at the molecule includes scattering from the nanoparticle, which is precisely the physics one often wants to keep in $\mu(t)$.

Dividing $\mu(\omega)$ by the *total* local field would remove plasmon-enhanced response. Dividing by a pure *vacuum incident* field $\mathbf{E}_{\text{inc}}$ removes only the source envelope, leaving nanoparticle / local-field physics inside μ .

5.2 Production runs (molecule + NP)

For each polarization i , the driver sets `plasmon.source.component = i`, and writes outputs under `x_dir/`, `y_dir/`, or `z_dir/`.

At every Meep timestep (and at $t = 0$):

1. sample $\mathbf{E}$ at the molecule site (`_get_electric_field`), converting Meep units to atomic units via `convertFieldMeep2Atomic`;
2. if $|\mathbf{E}|$ exceeds `tolerance_field_e`, call RT-TDDFT (`propagation`) to obtain μ ;
3. store μ for the next Meep source query (if back-coupling is on) and append both $\mathbf{E}$ and μ to CSV.

Additionally, an identical simulation begins but without the NP nor the molecule to calculate the reference E field spectra.

1. evaluates $\mathbf{E}(\mathbf{r}_{\text{mol}}, t)$ in atomic units;
2. appends (t, E_x, E_y, E_z) to the configured CSV.
3. once all three runs are done, the diagonal elements are merged into $\mathbf{E}_{\text{diag}}(t)$.

5.3 Unit and time-grid notes

Hybrid timesteps are conformed to Meep's Courant-limited Δt :

$$\Delta t_{\text{Meep}} = \frac{\text{Courant}}{\text{resolution}}, \quad \Delta t_{\text{a.u.}} = \Delta t_{\text{Meep}} \times \text{convertTimeMeep2Atomic}. \quad (10)$$

5.4 Spectral deconvolution

This subsection develops the hybrid Fourier formula from linear response, states precisely what is divided by what, and connects the continuous theory to the discrete operations in `fourier()`.

Solving for the polarizability is therefore a complex division,

$$\alpha_i(\omega) = \frac{\mu_i(\omega)}{E_i^{\text{drive}}(\omega)}, \quad (11)$$

whenever the denominator is non-vanishing. The absorption spectrum is built from $\text{Im } \alpha_i(\omega)$ (Section 3 and Eq. (2)), not from $\text{Im } \mu_i(\omega)$ alone: the latter still carries the spectral envelope of the Meep source.

If $E_i^{\text{drive}}(t) = \kappa \delta(t)$, then $E_i^{\text{drive}}(\omega) = \kappa$ (flat in frequency), so

$$\alpha_i(\omega) = \frac{\mu_i(\omega)}{\kappa} \propto \mu_i(\omega). \quad (12)$$

Hence the pure quantum path may take $\mathcal{R}_i(\omega) = S_{\mu,i}(\omega)$ up to a global scale that disappears after peak normalization. A Meep Gaussian (or other finite pulse) has $E_i^{\text{drive}}(\omega)$ that is neither flat nor real; omitting the division imprints the pulse spectrum and phase onto $A(\omega)$.

Write the electric field at the molecule site in the Full Plasmol run as

$$\mathbf{E}_{\text{loc}}(\mathbf{r}_{\text{mol}}, t) = \mathbf{E}_{\text{inc}}(t) + \mathbf{E}_{\text{scat}}(t), \quad (13)$$

where $\mathbf{E}_{\text{inc}}$ is the free-space incident field of the Meep source (same cell, source, and sample point, but no nanoparticle and no quantum molecule) and $\mathbf{E}_{\text{scat}}$ collects everything scattered by the nanoparticle (and, if back-coupling is on, radiation from the molecular dipole itself in later times).

Two natural choices for E_i^{drive} in Eq. (11) have very different meanings:

Denominator	What α_i measures
$E_{\text{loc},i}(\omega)$	Response relative to the <i>total</i> local field. Plasmonic near-field enhancement largely cancels between numerator and denominator; peaks track bare molecular resonances.
$E_{\text{inc},ii}(\omega)$	Response relative to the <i>vacuum incident</i> field. Enhancement and scattering remain inside $\mu_i(\omega)$; $A(\omega)$ reports the molecule as driven by the same free-space pulse that would be present without the NP.

PlasMol uses the second choice. Operationally, three vacuum Meep runs record $\mathbf{E}_{\text{inc}}^{(i)}(t)$ for $i \in \{x, y, z\}$; only the co-polarized diagonal is retained:

$$\mathbf{E}_{\text{diag}}(t) = \begin{pmatrix} E_x^{(x)}(t) \\ E_y^{(y)}(t) \\ E_z^{(z)}(t) \end{pmatrix} \equiv \begin{pmatrix} E_{\text{inc},xx}(t) \\ E_{\text{inc},yy}(t) \\ E_{\text{inc},zz}(t) \end{pmatrix}. \quad (14)$$

The production runs contribute the analogous diagonal dipole $\boldsymbol{\mu}_{\text{diag}}(t)$ from `fold`.

5.4.1 Consistent deconvolution

Define the discrete Fourier transforms used in code ($t_n = t_0 + n\Delta t$, $n = 0, \dots, N-1$):

$$S_{\mu,i}(\omega_k) = \Delta t \sum_{n=0}^{N-1} \mu_i(t_n) w(t_n) e^{-2\pi i kn/N}, \quad (15)$$

$$S_{E,i}(\omega_k) = \Delta t \sum_{n=0}^{N-1} E_{\text{inc},ii}(t_n) w(t_n) e^{-2\pi i kn/N}, \quad (16)$$

where $\omega_k = 2\pi \text{fttfreq}(N, \Delta t)_k$ in atomic units. Because the *same* window multiplies both time series, it factors in a manner compatible with Eq. (11): if $\mu = \chi * E_{\text{inc}}$ and one multiplies both signals by w before transforming, the ratio

$$\hat{\alpha}_i(\omega_k) := \frac{S_{\mu,i}(\omega_k)}{S_{E,i}(\omega_k)} \quad (17)$$

still estimates $\alpha_i(\omega_k)$ up to the usual finite-record and window-bias errors of a truncated FFT—it does *not* introduce an extra $w(\omega)$ factor in the numerator alone. (If only μ were windowed, the ratio would mix the polarizability with the Fourier transform of the window.)

5.4.2 Complex division, phase cancellation, and the amplitude floor

Write the complex spectra in polar form,

$$S_{\mu,i}(\omega) = |S_{\mu,i}| e^{i\phi_{\mu,i}}, \quad S_{E,i}(\omega) = |S_{E,i}| e^{i\phi_{E,i}}. \quad (18)$$

Then

$$\hat{\alpha}_i(\omega) = \frac{|S_{\mu,i}|}{|S_{E,i}|} e^{i(\phi_{\mu,i} - \phi_{E,i})}. \quad (19)$$

Two useful consequences follow immediately:

1. **Amplitude flattening.** Division by $|S_{E,i}|$ removes the Meep source envelope (e.g. the Gaussian spectrum set by `frequency` / `fwidth`). Residual structure in $|\hat{\alpha}_i|$ is molecular and/or plasmon-mediated.
2. **Phase cancellation.** The relative phase $\phi_{\mu,i} - \phi_{E,i}$ is the phase lag of the induced dipole relative to the vacuum incident field. Propagation delay from the Meep source plane to $\mathbf{r}_{\text{mol}}$, and the overall complex scale of the source, largely cancel. Absorption depends on $\text{Im } \hat{\alpha}_i$, which is the component of the response in phase quadrature with E_{inc} .

In components,

$$\text{Re } \hat{\alpha}_i = \frac{\text{Re } S_{\mu,i} \text{ Re } S_{E,i} + \text{Im } S_{\mu,i} \text{ Im } S_{E,i}}{|S_{E,i}|^2}, \quad (20)$$

$$\text{Im } \hat{\alpha}_i = \frac{\text{Im } S_{\mu,i} \text{ Re } S_{E,i} - \text{Re } S_{\mu,i} \text{ Im } S_{E,i}}{|S_{E,i}|^2}. \quad (21)$$

Equation (21) is the quantity summed into $A(\omega)$. The code stores both real and imaginary parts (`abs_real`, `abs_imag` in the optional NPZ); only the imaginary part enters `absorption()`.

Near frequencies where the vacuum pulse has little power, $|S_{E,i}| \rightarrow 0$ and Eq. (17) is ill-conditioned. PlasMol therefore applies a relative amplitude floor on the spectral window of interest only. Let

$$\mathcal{W} = \{ k : \text{min_ev} \leq \hbar\omega_k \leq \text{max_ev} \} \quad (22)$$

(with $\hbar\omega$ converted to eV as in Section 3), and

$$E_{\text{max},i} = \max_{k \in \mathcal{W}} |S_{E,i}(\omega_k)|, \quad \varepsilon = \varepsilon_{\text{rel}} E_{\text{max},i}, \quad \varepsilon_{\text{rel}} = 10^{-8} \text{ (e_floor_rel)}. \quad (23)$$

The regularized estimator is

$$\alpha_i(\omega_k) = \begin{cases} \frac{S_{\mu,i}(\omega_k)}{S_{E,i}(\omega_k)} & \text{if } k \in \mathcal{W} \text{ and } |S_{E,i}(\omega_k)| > \varepsilon, \\ 0 & \text{otherwise.} \end{cases} \quad (24)$$

If no bin in $\mathcal{W}$ exceeds the floor for a given polarization, that direction contributes nothing and a warning is logged. In practice this means: either widen the Meep source bandwidth so that $|E_{\text{inc}}(\omega)|$ covers $[\text{min_ev}, \text{max_ev}]$, or shrink the plot window to the support of the pulse.

5.4.3 From α_i to the reported absorption

With $\mathcal{R}_i(\omega) := \alpha_i(\omega)$ in the hybrid path, Eq. (2) becomes

$$A(\omega) = -\frac{4\pi\omega}{3c} \sum_{i \in \{x,y,z\}} \text{Im } \alpha_i(\omega). \quad (25)$$

For an isotropic molecule in free space one recovers the familiar orientational average of the dipole polarizability, $\frac{1}{3} \sum_i \text{Im } \alpha_{ii}$. With a nanoparticle the same algebraic form is retained, but each α_i is an *effective* polarizability of the molecule as driven by the vacuum incident field in that geometry—not the bare gas-phase α .

Finally the curve is peak-normalized for shape comparison,

$$A_{\text{norm}}(\omega) = \frac{A(\omega)}{\max_{\omega \in \mathcal{W}} |A(\omega)|}, \quad (26)$$

so overall scales (kick strength, Meep amplitude, unit conventions shared by μ and E_{inc}) drop out, provided they are consistent between numerator and denominator.

5.4.4 Compact algorithm (as implemented)

1. Align $\mu_{\text{diag}}(t)$ and $\mathbf{E}_{\text{diag}}(t)$ on a common length and Δt (`_align_series`).
2. Form $w(t)$ from γ (and τ if set).
3. For each $i \in \{x, y, z\}$: compute $S_{\mu,i}$ and $S_{E,i}$ by Eqs. (15)–(16); restrict to $\mathcal{W}$; apply Eq. (24).
4. Assemble $A(\omega)$ from Eq. (25); normalize; write PNG/CSV and optional NPZ (`deconvolved=True`).

Summary of interpretation.

- Division by vacuum $E_{\text{inc}}(\omega)$ flattens the Meep source spectrum and cancels its phase (Eqs. (19)–(21)).
- Nanoparticle-induced local fields remain inside $\mu(\omega)$, so plasmon-enhanced features are retained (Eq. (13) and the table above).
- Bins with negligible $|E_{\text{inc}}|$ are zeroed rather than producing numerical blow-ups (Eq. (24)).

5.5 Example hybrid JSON (Fourier + Au NP)

```
{
  "settings": {
    "dt": 0.1,
    "t_end": 10000,
    "driver": "fourier"
  },
  "plasmon": {
    "simulation": {
      "cell_length": 0.2,
      "pml_thickness": 0.05
    },
    "source": {
      "type": "gaussian",
      "center": [-0.04, 0, 0],
      "size": [0, 0.2, 0.2],
```



```

    "component": "z",
    "is_integrated": true,
    "additional_parameters": {
      "frequency": 2.291666667,
      "fwidth": 2.083333335
    }
  },
  "nanoparticle": {
    "material": "Au",
    "radius": 0.025
  },
  "molecule": {
    "position": [0, 0, 0.026451]
  }
},
"molecule": {
  "geometry": "Na.xyz",
  "geometry_units": "bohr",
  "charge": 0,
  "spin": 1,
  "basis": "631g*",
  "xc": "pbe0",
  "hermiticity_tolerance": 1e-12,
},
"files": {
  "field_e_filepath": "field_e.csv",
  "field_p_filepath": "field_p.csv",
  "spectra_e_vs_p_filepath": "output.png"
},
"additional_parameters": {
  "fourier": {
    "spectrum_filepath": "spectrum.png",
    "min_ev": 1.5,
    "max_ev": 5.0,
    "field_e_ref_filepath": "field_e_ref.csv"
  }
}
}

```